

Machine Learning Models for Mental-Health Prediction from Actigraphy: A Systematic Appraisal and an Empirical Demonstration

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Highlights.

- Systematic PROBAST+AI/TRIPOD+AI appraisal of 28 published models on open actigraphy cohorts
- Day-level cross-validation inflated AUROC by up to 0.17 and created signal where there was none
- Frozen cross-cohort transfer lost up to 0.21 AUROC with collapse of calibration
- No published model on these cohorts reports calibration; honest slopes were 0.53 to 0.95
- Subject-wise, calibrated baselines with public code and run manifests for future comparison

Abstract. Background and objective. The open actigraphy cohorts DEPRESJON (depression), PSYKOSE (schizophrenia) and HYPERAKTIV (ADHD) are the most used public benchmarks for machine-learning detection of mental disorders from wearables, and reported accuracies have risen from the dataset authors' leave-one-patient-out 0.7 to above 0.9. We asked whether this reflects better models or weaker evaluation. **Methods.** Part I appraises every published supervised model on the cohorts with PROBAST+AI and TRIPOD+AI, recording the unit of train/test splitting, external validation, calibration reporting, events per predictor, code availability and handling of the control group shared by DEPRESJON and PSYKOSE. Part II re-evaluates the cohorts with models fixed a priori, ten seeds and participant-level bootstrap intervals: record-wise versus subject-wise cross-validation (E1), frozen cross-cohort transfer (E2), calibration and recalibration (E3), sex strata (E4), transdiagnostic and five-class tasks (E5) and ablation (E6). **Results.** Of 28 published models, 7 split by participant, 9 by day and 12 did not state the unit; none reported calibration or validated on a second cohort. Record-wise splitting inflated AUROC by 0.065 to 0.114 (DEPRESJON), 0.020 to 0.044 (PSYKOSE) and 0.130 to 0.169 (HYPERAKTIV), where the honest estimate was at chance. Models trained on PSYKOSE lost 0.13 to 0.21 AUROC on DEPRESJON with calibration slopes of 0.27 to 0.63; honest slopes were 0.53 to 0.95, and recalibration did not repair the transfer. Honest performance was stable under ablation. **Conclusions.** Reported progress on these benchmarks is largely an artefact of evaluation design. We provide subject-wise, calibrated, cross-cohort baselines with public code against which future claims can be checked.

Keywords. actigraphy; depression; schizophrenia; ADHD; DEPRESJON; PSYKOSE; HYPERAKTIV; OBF-Psychiatric; data leakage; external validation; calibration; PROBAST+AI; TRIPOD+AI; reproducibility.

Declaration of competing interest. The author declares that she has no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Ethics. The study is a secondary analysis of publicly available, de-identified datasets (DEPRESJON, PSYKOSE, HYPERAKTIV, OBF-Psychiatric) released for research use by their custodians; no new data were collected and no ethical approval was required.

Data availability. The datasets are available from their custodians: DEPRESJON (<https://datasets.simula.no/depresjon/>), PSYKOSE (<https://datasets.simula.no/psykose/>), HYPERAKTIV (<https://datasets.simula.no/hyperaktiv/>) and OBF-Psychiatric (<https://zenodo.org/records/13754984>). SHA-256 checksums of every file used are recorded in the code repository.

Code availability. All analysis code, configuration files, run manifests, per-window predictions and the scripts that generate every table and figure are available at <https://github.com/rogerpanel/cv/tree/main/mlmh-appraisal> (release v1.0.0-submission; Zenodo DOI to be added on acceptance). An anonymised copy is provided to the editor for review.

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Declaration of generative AI in the writing process. During the preparation of this work the author used an AI coding assistant (Claude Code, Anthropic) to scaffold the analysis code and to draft text under the author’s direction. After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the content of the publication.

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